(A) Data manipulation can also be done with indexing.

Reading from sources

(1) from json

```
>>> import pandas as pd
    from ~~import~~ io import StringIO

    Data = {' . . . . . '}

>>> df = pd.read_json(StringIO(Data))
```

↳ convert into string
↳ string i/p o/p.

Now, to convert back,

```
>>> df.to_json()
    df.to_json(orient='index')
```

X — same — X

default

∴ `df.to_json(orient='records')`

↳ json conversion based on records.

~~# From csv url.~~

~~>>> df.read_csv~~

PTO

(A) During data manipulation, can also use index

(A) To keep a Df a df. → use double ([]) during data manipulation for single column

② From csv url (comma separated values)

```
>>> df = pd.read_csv('url', header='None')
```

pandas — read_csv — if no header in data

To convert back,

```
>>> df.to_csv('file_name')
```

saves in same directory.

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③ From html

! pip install

3 ~~libra~~ packages ~~need~~ needed: lxml, html5lib, beautifulsoup4

```
>>> df = pd.read_html('url')
```

→ Creates a list of tables ✓✓

⇒ ```
>>> df[0]
```
✓ → for 1st element of the list

(*)(#) To match by some keyword,

```
>>> df = pd.read_html('url', match='country', header=~~None~~0)
```

matches for country. — takes 1st row as header

```
>>> df[0]
```
```

(4) From Excel (doc.xlsx)

```
df = pd.read_excel('file.xlsx')
```

Note → if multiple sheets,

```
df = pd.read_excel(file, sheet_name='...')
```

by default, 1st sheet

```
! pip install *openpyxl
```

→ we can convert df to pickle file

```
>>> df.to_pickle('df_excel')
```

↳ cannot open this file

```
>>> df = pd.read_pickle(p_file)
```

↳ to read from pickle file.

Note: pickle file → converts a python object into bytestream & stores it

→ maintains program state accross sessions.